Fiscal Year:	FY 2005	**Task Last Updated:**	FY 03/15/2005
PI Name:	Tash, Joseph S Ph.D.		
Project Title:	Negative Impacts of Altered Gravity Models on Male Mammalian Reproductive Capacity		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Developmental biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Reproductive Biology (2) Developmental Biology		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
PI Email:	jtash@kumc.edu	**Fax:**	FY 913-588-7180
PI Organization Type:	UNIVERSITY	**Phone:**	913-526-4103
Organization Name:	University of Kansas Medical Center		
PI Address 1:	Department of Molecular and Integrative Physiology		
PI Address 2:	3901 Rainbow Boulevard		
PI Web Page:			
City:	Kansas City	**State:**	KS
Zip Code:	66160-0001	**Congressional District:**	3
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2003 Space Biology 03-OBPR-03
Start Date:	03/01/2004	**End Date:**	09/30/2006
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Ronca, April E. Ph.D. (Wake Forest University School of Medicine) Wade, Charles E. (US Army Institute of Surgical Research)		
Grant/Contract No.:	NNA04CC54A		
Performance Goal No.:			
Performance Goal Text:			
	The overall hypothesis driving the proposed research is that extended exposure to microgravity (μG) has a negative impact on the capacity of males of mammalian species to reproduce. This hypothesis is based on previous published work from our laboratory, demonstrating that hind limb sus-pension (HLS) of adult male rats for 6 weeks causes a block of spermatogenesis leading to permanent sterility. Furthermore, the initial 6 week HLS treatment triggers a decline in testicular function that continues beyond the initial HLS treatment period resulting in massive apoptosis of the germinal epithelium and sterility when assessed 6 mos later. One of the initial insults to the testis caused by HLS is a reduction in blood flow to levels just above the minimum known to sustain spermatogenesis. Essential components of this productive experimental approach have been to utilize i) partial inguinal to prevent the testes from becoming abdominal during hind limb elevation, and ii) pair-fed control animals identical to the HLS animals with inguinal ligation and tail harnessing,		

Task Description:	but elevated only to lift the tail but not limit contact of the hind limbs with the cage floor. New goals raised by the results during the previous project period are 1. Determine the maximum duration of HLS that permits subsequent restoration of spermatogenesis and fertility. 2. Ascertain the role that compromised blood flow and reactive oxygen species (ROS) play in the mechanisms that block spermatogenesis in long term HLS, and whether dietary administration of inhibitors of ROS damage serve as a countermeasure to the loss of spermatogenesis 3. Determine if prepubertal testicular development and adult testicular function are compromised in hypergravity (HG). This new program is a logical extension of the studies currently underway as part of our current NASA grant and represent new efforts directly and unambiguously related to the top two questions that the Office of Biological and Physical Sciences supported research must answer: a) How can we assure the survival of humans traveling far from Earth? and 2) How does space change life forms; how may humankind flourish? The proposed HG studies represent a new collaboration between two experienced space flight and ground based research teams at NASA Ames Research Center and the University of Kansas Medical Center. Given the progress already achieved in the first 2 years of the current grant we expect to continue to provide answers to these important questions. Furthermore, the new studies will begin to test potential countermeasures for likely causes for the loss of spermatogenesis in the HLS model, and embark on novel studies on the effect of HG on testicular development as initial supporting data in preparation for eventual studies on testicular function in long term µG. With regard to the new Exploration Mission directive from the President for NASA regarding extended manned missions to the moon and Mars, the Bioastronautics Critical Path Roadmap (BCPR) risks include “unacceptable levels of increased hereditary, fertility, or sterility risk caused by occupational radiation exposure or the combined effects of radiation and other space flight factors.” In this regard, previously published short-term µG studies in astronauts and rats suggest that testicular function may be compromised in longer duration space flight. However, male reproductive health data for either species following long term µG has yet to be collected and analyzed. Using hindlimb unloading (HLS) as a animal model for µG, we previously demonstrated that 6 wks of HLS in sexually mature male rats leads severely inhibited spermatogenesis. In relation to the BCPR, this can cause significantly increased health risks not only for male infertility or sterility, but also for increased testicular cancer due to the well known combined effect of testicular hyperthermia and radiation exposure. Ultimately, this model will have to be verified by long term rodent and human space flight experiments, developing appropriate countermeasures, and monitoring male astronaut health post flight.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Male infertility represents approximately half of the cases of infertility in couples. Inflammatory responses to hyperthermia and/or altered blood flow are a common component of male infertility of a broad spectrum of etiologies. The long term hindlimb unloading model appears to reproduce many of the same alterations in testis histopathology associated with many male infertility scenarios. Our studies in relation tp specific aim 3 in particular will study the efficicacy of potential countermeasures for inflammatory and post-iscemic induced male infertity regardless of whether in microgravity or in earth normal gravity.
Task Progress:	The Bioastronautics Critical Path Roadmap (BCPR) risks include “unacceptable levels of increased hereditary, fertility, or sterility risk caused by occupational radiation exposure or the combined effects of radiation and other space flight factors.” In this regard, previously published short-term µG studies in astronauts and rats suggest that testicular function may be compromised in longer duration space flight. However, male reproductive health data for either species following long term µG has yet to be collected and analyzed. Us¬ing hindlimb unloading (HLS) as a animal model for µG, we previously demonstrated that 6 wks of HLS in sexually mature male rats leads severely inhibited spermatogenesis [1]. Even after a 6 month recovery period, animals showed irreversible aspermatogenic sterility similar to testicular regression syndrome, even though the testes are prevented from becoming abdominal by partial ligation of the inguinal canal [2]. The present study was undertaken to determine if the hindlimb unloaded animal model produces changes in testicular morphometry and/or scrotal temperature that could account for the generation of aspermatogenic sterility. The location of the testis during HLS was examined directly using MRIs performed on a 9.4 T horizontal bore Varian Scanner. On a weekly basis, rats were removed from the HLS cages and positioned in a specially designed Plexiglas cradle maintaining suspension of the hindlimbs from tails and also housed within a 14 cm diameter radio frequency coil. In this design, the head downward tilt was maintained in the coil using a foam plastic insert and tape to retain the tail-up position used in the HLS cages. MRIs were acquired in 30 contiguous axial and sagittal planes using spin echo sequence with parameters TR/TE=2000/12 ms, field-of-view= 70mm X 70mm for axial and 70mm X 90mm for sagittal, image ma¬trix=128X128, slice thickness=2mm and number-of-averages=2. Infrared thermography was performed using a FLIR Thermovision A40M camera system and software. The data analyzed in 2D and 3D confirmed that partial suturing of the inguinal canal using a purse-string stitch, as performed previously, prevents the testes from descending into the abdomen during HLS treat¬ment. Significant differences in testis shape and position against the body were observed in HLS versus normal horizontally positioned animals. In the head downward HLS position, the testes, while remaining outside the abdomen and scrotal, were less elongated and pressed against the dorsal end of the abdomen and hindlimbs, thus reducing the surface areas available for cooling the testis below body temperature. Indeed, infrared thermography confirmed that scrotal temperatures were significantly higher (2.2 ± 0.6°C) than free roaming controls ($P<0.00002$). This temperature is almost double the increase that occurs acutely in humans during normal horizontal bed rest. Maintenance of testicular temperature well below body temperature is critical for safeguarding spermatogenesis. The primary mechanism for this regulation is testicular position relative to the body and is determined by a balance between the outward pull of gravity and the inward pull of the cremasteric muscles. Thus, we hypothesize that in long term space flight, chronic testicular hyperthermia may occur in both rats and humans. In relation to the BCPR, this can cause significantly increased health risks not only for male infertility or sterility, but also for increased testicular cancer due to the well known combined effect of testicular hyperthermia and radiation exposure. Ultimately, this model will have to be verified by long term rodent and human space flight experiments, developing appropriate countermeasures, and monitoring male astronaut health post flight
Bibliography Type:	Description: (Last Updated: 01/24/2022)

Papers from Meeting Proceedings	Tash, J.S.; Wolfe, S.; Timmerburg, B; Bilgen, M. "MAGNETIC RESONANCE IMAGING (MRI) AND IR THERMOGRAPHIC ANALYSIS OF TESTICULAR FEATURES UNDERLYING STERILITY IN LONG-TERM HINDLIMB UNLOADED (HLS) MALE RATS" Bioastronautics Investigators' Workshop, Galveston, TX Jan-2005